



OPPORTUNITIES FOR THE BIGGEST JOB ON EARTH

BASF – Purdue Data Mine Wine Varietals Forecast

Fall 2024

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The Different Segments of BASF



Chemicals

Petrochemicals
Intermediates



Materials

Performance Materials
Monomers



Industrial Solutions

Dispersions & Resins
Performance Chemicals



Surface Technologies

Catalysts
Coatings



Nutrition & Care

Care Chemicals
Nutrition & Health



Agricultural Solutions

Project falls within:

SOLUTIONS FOR THE
BIGGEST JOB
ON EARTH

Internal

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BASF Agricultural Solutions Portfolio

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Field crop seeds

Fully enabled seeds and traits businesses for canola/oilseed rape, soybean and cotton



Vegetable seeds

Long-standing, successful global vegetable seeds business under the Nunhems® brand



Seed treatment

New seed treatment portfolio including nematicide seed treatment and research projects



Crop protection

Innovative crop protection products designed to meet customer needs and demands



Herbicides

Non-selective herbicides complements BASF's existing selective herbicide portfolio



Research platforms

Hybrid wheat, juncea (Indian mustard) and other projects to strengthen our innovation pipeline



Digital farming solutions

xarvio™ – global digital farming products and pipeline developments



Traits portfolio

Unparalleled gene discovery platform focusing on yield and quality traits

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Internal

What will farming look like in 2030?

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**50% higher
productivity
required**

Drives growth
in high-quality
seeds and crop
protection

**Sustainability
requirements**

Biodiversity
CO₂ neutral farming
Transparency

**Pest
and disease
resistance**

Strong demand
for novel modes
of action

**Increased farm
professionalization**

Strong focus on
digitalization and
farm management
systems

About BASF Agricultural Solutions

SOLUTIONS FOR THE
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When discussing wine, industry experts often utilize growing years as a key characteristic of wine varieties

- Wine varieties are also discussed in terms of years ([link](#))
- Key elements (bioclimatic indices)
 - Estimated rainfall
 - Estimated groundwater availability / drought monitor
 - Estimated temperature & temp variations
 - Estimated Growing Degree Days (GDUs)
 - Other agronomic factors (planting dates, trendline crop yields, harvest dates)

Tools/Skills Learned: Data exploration, data integration machine learning, time series analysis, Python, AI forecast generation

Keywords: Machine learning, weather, agriculture, viticulture, climate

Project Request

- Build off the previous Data Mine team and their LSTM model to determine which weather variables within the model have the highest relationship with yield
- Investigate AI driven long-range weather forecast models (ideally 3 + month) to predict next season grape wine yields for the identified California Counties
- Create a digital means for interacting with the platform (i.e. why is 2021 like 1981; why 2024 will likely be like 1998)

BASF – Purdue Data Mine Proposal

Digitizing / Forecasting Analogue Years for 10 Key Crops

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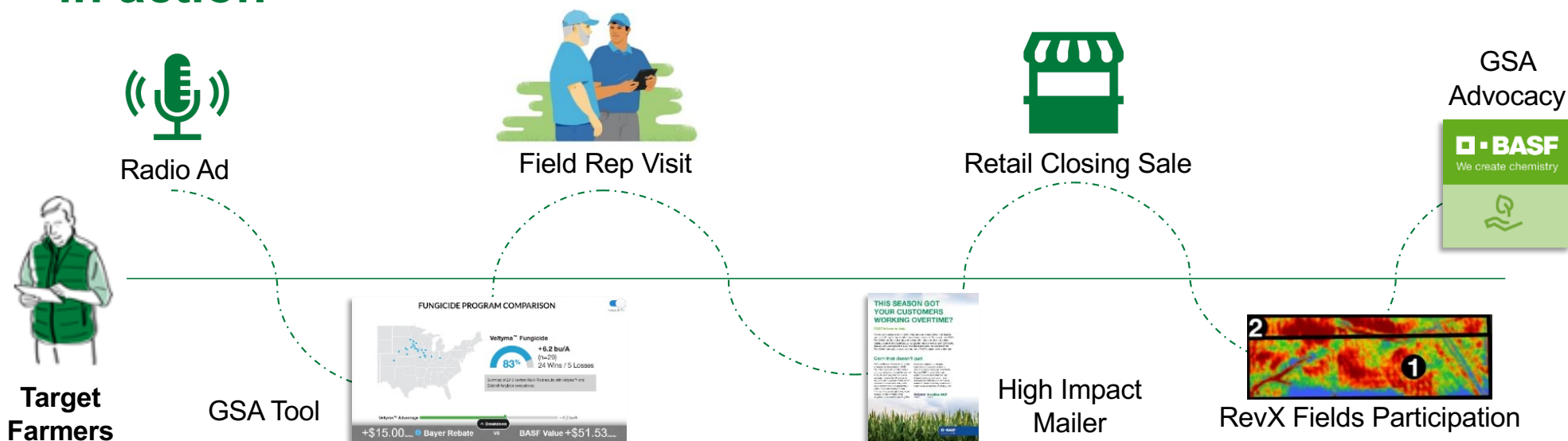
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Applications – Utilizing 2024 Crop Outlook in action

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Target Customers

- Key retail organization for business planning and co-marketing
- Agriculture industry associations for ag-vocacy work
- Agriculture research institutions working on climate change, food security/insecurity and other issues

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